

SOLID Principles

- S - Single responsibility principle (SRP)
- O - Open Closed Principle (OCP)
- L - Liskov substitution Principle (LSP)
- I - Interface segregation Principle (ISP)
- D - Dependency inversion Principle (DIP)

```
1 public class Employee {  
2     private int id; // Employee ID  
3     private String name; // Employee Name  
4     private String address; // Employee Address  
5  
6     public Employee(int id, String name, String address) {  
7         this.id = id;  
8         this.name = name;  
9         this.address = address;  
10    }  
11  
12    public void printPerformanceReport() {  
13        // Code to print performance report  
14        System.out.println("Performance report of employee: " + name);  
15    }  
16  
17    public double computeSalary() {  
18        // Code to compute salary  
19        return 1000.0;  
20    }  
21  
22    public void updateEmployeeData() {  
23        // Code to update employee data  
24        System.out.println("Employee data updated successfully");  
25    }  
26  
27    public void fetchEmployeeData() {  
28        // Code to fetch employee data  
29        System.out.println("Employee data fetched successfully");  
30    }  
31 }
```

If the report format changes,
we might need to update this function.

This class is trying to do
too many things.
k

Say, if the taxation changes, we might
need to update this function.

If data storage requirements change,
we might need to update this function

Because the above class is doing too many things, there are too many reasons to update the code present in the class. This violates Single responsibility principle.

Single responsibility principle states that, there should one and only one reason to change a piece of code.

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```
3 public class Employee {  
4     private int id; // Employee ID  
5     private String name; // Employee Name  
6     private String address; // Employee Address  
7  
8     public Employee(int id, String name, String address) {  
9         this.id = id;  
10        this.name = name;  
11    }
```

```
11        this.address = address;  
12    }  
13  
14    public int getEmployeeId() {  
15        return id;  
16    }  
17  
18    public String getEmployeeName() {  
19        return name;  
20    }  
21  
22    public String getEmployeeAddress() {  
23        return address;  
24    }  
25  
26    public void setEmployeeAddress(String address) {  
27        this.address = address;  
28    }  
29  
30    public void setEmployeeName(String name) {  
31        this.name = name;  
32    }  
33 }
```

→ This class is now only responsible for
Basic employee data creation and fetching

```
32 }  
33 }  
  
public class EmployeeSalaryCalculator {  
    public double computeSalary(Employee e) {  
        // Code to compute salary  
        return 1000.0;  
    }  
}
```

This class is only responsible
for computing salary

```
3 public class EmployeePerformanceReportGenerator {  
4     public void printPerformanceReport(Employee e) {  
5         // Code to print performance report  
6         System.out.println("Performance report of employee: " + e.getEmployeeName());  
7     }  
8 }  
9
```

This class is only responsible for
printing performance report

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We have segregated the classes in a way such that their is one core responsibility every class has, hence there is only one reason to change logic in a class.

OCP - Open closed principle

A class should be open for extensions but closed for modifications.

OCP - Open closed principle

A class should be open for extensions but closed for modifications.

```
public class NotificationSender {  
    public void sendNotifications(List<String> notificationTypes, String message) {  
        for (String notificationType : notificationTypes) {  
            switch (notificationType) {  
                case "EMAIL":  
                    EmailNotification notification = new EmailNotification();  
                    notification.sendEmailNotification(message);  
                    break;  
                case "SMS":  
                    SMSNotification smsNotification = new SMSNotification();  
                    smsNotification.sendSMSNotification(message);  
                    break;  
                case "PUSH":  
                    PushNotification pushNotification = new PushNotification();  
                    pushNotification.sendPushNotification(message);  
                    break;  
            }  
        }  
    }  
}
```

→ This class will be modified every time we introduce or remove any type of notification.

This is a clear violation of OCP

How to improve? We should try to think that if further alteration in the requirements can unnecessary impact code of a class, it is violating OCP

```
3 public interface Notification {  
4  
5     void sendMessage(String message);  
6 }  
7
```

→ Interface to represent any notification

```
3 public interface Notification {  
4  
5     void sendMessage(String message);  
6  
7 }
```

Interface to represent any notification

Individual type of notification implementing the interface

```
2  
3 public class EmailNotification implements Notification {  
4  
5     @Override  
6     public void sendMessage(String message) {  
7         System.out.println("Email notification: " + message);  
8     }  
9  
10 }
```

```
3 public class PushNotification implements Notification {  
4  
5     @Override  
6     public void sendMessage(String message) {  
7         System.out.println("Push notification: " + message);  
8     }  
9  
10 }
```

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